

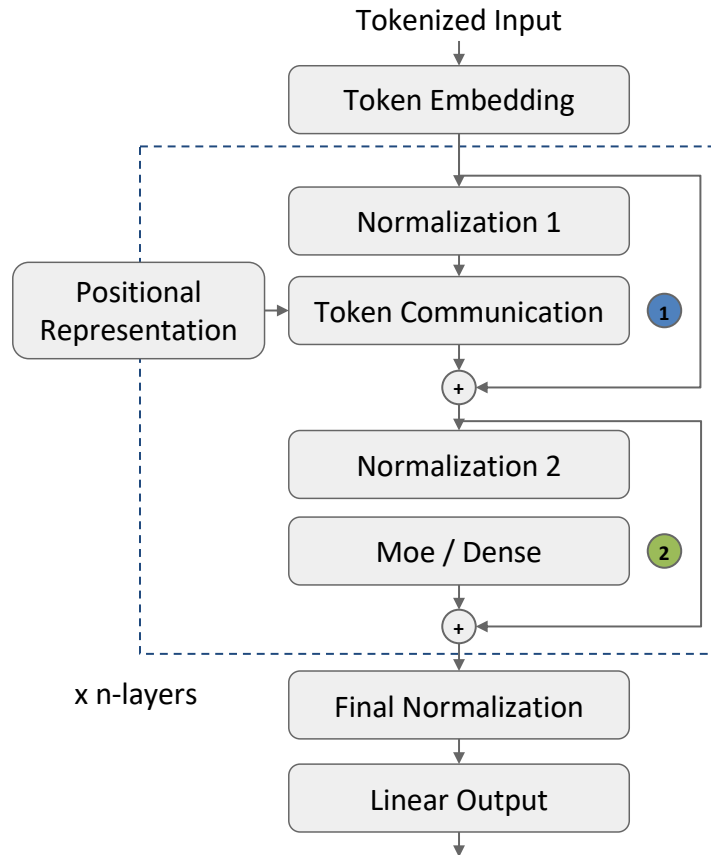
LLM Aggregate

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Architectures



1

State Space Models (SSM)		Updates a state that compresses information, making it more in long-context scenarios.
Attention (Softmax)	Casual	Tokens can only attend to itself and previous tokens.
	Sliding Window	Limits what tokens can attend the current token, faster computational time, loss of information.
	Full	Every token can attend to everyone.

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MoE	Only subset of multiple dense blocks are used, increasing model capacity, with less computational time.
Dense	All parameters are used, easier to fine-tune.



Architectures

KV Cache Methods	Reduce KV Cache Size	Cache Size (descending)
Multi-Head Attention (MHA) (Baseline)	Each attention head has its own query, key, value.	Largest
Group-Query Attention (GQA)	Groups of queries share keys and values to reduce memory.	
Multi-Query Attention (MQA)	All queries share keys and values.	
Multi-Head Latent Attention (MLA)	Stores keys and values in a compressed form, and reconstructing when needed.	Smallest

Positional Representation	Embed positional information of words.
Rotary Positional Embeddings (RoPE)	Rotates vector embeddings according to relative positions.
ALiBi	Bias attention scores so closer tokens are preferred and farther are penalized.
Learned Absolute	Separate positional vector for each absolute positions that is added to the token embeddings.



Meta: Llama

Model		Release Date	Parameters	Active	Architecture	Context
Llama 1		Feb 2023	7B/13B/33B/65B	—	Dense	2k
Llama 2		Jul 2023	7B/13B/34B/70B	—	Dense	4k
Llama 3		Apr 2024	8B/70B	—	Dense	8k
Llama 3.1		Jul 2024	8B/70B/405B	—	Dense	128k
Llama 3.2		Sep 2024	1B/3B/11B/90B	—	Dense	128k
Llama 3.3		Dec 2024	70B	—	Dense	128k
Llama 4	Scout	Apr 2025	109B total	17B	MoE (16)	10m
	Maverick	Apr 2025	400B total	17B	MoE (128)	1m



IBM: Granite

Model		Release Date	Parameters	Active	Architecture	Context
Granite 3.0		Oct 2024	2B/8B	—	Dense	128K
			1B/3B	400M/800M	MoE	128K
Granite 3.1		Dec 2024	2B/8B	—	Dense	128K
			1B/3B	400M/800M	MoE	128K
Granite 3.2		Feb 2025	2B/8B	—	Dense	128K
Granite 3.3		Apr 2025	2B/8B	—	Dense	128K
Granite 4.0	Small	Oct 2025	32B	9B	MoE + SSM	>128K
	Tiny		7B	1B	MoE + SSM	>128K
	Micro		1B	—	Dense + SSM	>128K
Granite 4.1		Apr 2026	3B/8B/30B	—	Dense	131K (Instruct) / 512K (Base)



Google: Gemma

Model	Release Date	Parameters	Active	Architecture	Context
Gemma 1	Feb 2024	2B/7B	—	Dense	2k
Gemma 2	Jun 2024	2B/9B/27B	—	Dense	4k
Gemma 3	Mar 2025	1B/4B/12B/27B	—	Dense	8k
Gemma 3n	Jul 2025	5.44B	1.91B	Dense+PLE	128k
Gemma 4	Apr 2026	5.1B	2.3B	Dense+PLE	128k
		8B	4.5B	Dense+PLE	128k
		25.2B	3.8B	MoE	256k
		30.7B	—	Dense	256k



Mistral

Model	Release Date	Parameters	Active	Architecture	Context
Mistral 7B	Sep 2023	7B	—	Dense	32K
Mixtral 8x7B	Dec 2023	46.7B total	12.9B	MoE	32K
Mixtral 8x22B	Apr 2024	141B total	39B	MoE	65K
Mistral Large 2	Jul 2024	123B	—	Dense	128K
Mistral Small 3	Jan 2025	24B	—	Dense	128K
Ministral 3	Dec 2025	3B/8B/14B	—	Dense	256K
Mistral Large 3	Dec 2025	675B	41B	MoE	256K
Mistral Small 4	Mar 2026	119B	6.5B	MoE	256K
Mistral Medium 3.5	Apr 2026	128B	—	Dense	256K



Qwen

Model	Release Date	Parameters	Active	Architecture	Context
Qwen 1.5	Feb 2024	0.5B/1.8B/4B/7B/14B/32B/72B/110B	—	Dense	32K
Qwen 2	June 2024	0.5B/1.5B/7B/72B	—	Dense	128K
		57B	14B	MoE	128K
Qwen 2.5	Sep 2024	0.5B/1.5B/3B/7B/14B/32B/72B	—	Dense	128K
Qwen 3	Apr 2025	0.6B/1.7B/4B/8B/14B/32B	—	Dense	128K
		30B/235B	3B/22B	MoE	128K
Qwen 3.5	Feb 2026	0.8B/2B/4B/9B/27B	—	Dense	262K
		35B/122B/397B	3B/10B/12B	MoE	262K
Qwen 3.6	April 2026	35B	3B	MoE	262K
		27B	—	Dense	262K



OpenAI: GPT-OSS

Model	Release Date	Parameters	Active	Architecture	Context
gpt-oss-20b	Aug 2025	20.9B total	3.6B	MoE	128K
gpt-oss-120b	Aug 2025	116.8B total	5.1B	MoE	128K



DeepSeek

Model		Release Date	Parameters	Active	Architecture	Context
DeepSeek V1		Jan 2024	7B/67B	—	Dense	4K
DeepSeek V2		May 2024	236B	21B	MoE	128K
DeepSeek V3		Dec 2024	671B	37B	MoE	128K
DeepSeek R1		Jan 2025	671B	37B	MoE	128K
DeepSeek V4	Pro	Apr 2026	1.6T	49B	MoE	1M
	Flash		284B	13B	MoE	1M



MiniMax

Model	Release Date	Parameters	Active	Architecture	Context
MiniMax-Text-01	Jan 2025	456B total	45.9B	MoE + Lighting Attention	1M
MiniMax-VL-01	Jan 2025	456B total	45.9B	MoE	1M
MiniMax-M1	Jun 2025	456B total	45.9B	MoE + Lighting Attention	1M
MiniMax-M2.1	Dec 2025	229B	10B	MoE	205K
MiniMax-M2.5	Feb 2026	229B	10B	MoE	205K
MiniMax-M2.7	May 2026	229B	10B	MoE	205K
MiniMax-M3	Jun 2026	428B	23B	MoE	1M



GLM

Model	Release Date	Parameters	Active	Architecture	Context
ChatGLM 6B	2022–2023	6B	—	Dense	2K
GLM-4 9B	Jun 2024	9B	—	Dense	128K
GLM-4-32B	Apr 2025	32B	—	Dense	128K
GLM-4.5	Jul 2025	355B total	32B	MoE	200K
GLM-4.6	Sep 2025	355B total	32B	MoE	200K
GLM-4.7	Dec 2025	355B total	32B	MoE	200K
GLM-5	Feb 2026	744B total	40B	MoE	200K
GLM-5.1	Apr 2026	754B total	40B	MoE	200K
GLM-5.2	Jun 2026	744B	40B	MoE	1M



Moonshot: Kimi

Model	Release Date	Parameters	Active	Architecture	Context
Kimi K2	Jul 2025	1T total	32B	MoE	1T total
K2-Instruct	Sep 2025	1T total	32B	MoE	1T total
K2-Thinking	Nov 2025	1T total	32B	MoE	1T total
Kimi K2.5	Jan 2026	1T total	32B	MoE	1T total
Kimi K2.6	Apr 2026	1T total	32B	MoE	1T total
Kimi K2.7 Code	Jun 2026	1T total	32B	MoE	1T total